16 Solve the simultaneous equations

$$3x + y = -4$$
$$3x - 4y = 6$$

Subtract the second equation from the first.

$$\begin{array}{rrcl} & 3x + y & = & -4 \\ - & 3x - 4y & = & 6 \\ \hline & 5y & = & -10 \\ & \div 5 & & \div 5 \\ & y & = & -2 \end{array}$$

1 mark
Eliminate x by subtraction

Substitute $y = -2$ into $3x + y = -4$.

$$\begin{array}{rcl} 3x - 2 & = & -4 \\ +2 & & +2 \\ 3x & = & -2 \\ \div 3 & & \div 3 \\ x = \dfrac{-2}{3} & & \end{array}$$

1 mark
Substitute to find x

$x = \dfrac{-2}{3}$

$y = -2$

1 mark
Both values correct

(Total for Question 16 is 3 marks)